



Data Mining I

S1

NOVA-IMS 2025/2026

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Agenda

Lec. 1	8 Sep	Course presentation (Syllabus) - Syllabus - Requirements - Practical Sessions - Evaluation Method - Course project Introduction to Data Science - Definitions - Data Science roles and skills - Fundamental principles of data-driven thinking The canonical tasks in Data Mining and work process - Canonical tasks in Data Mining - Supervised learning - Unsupervised learning - Data Science application examples
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Syllabus



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DATA MINING I

SYLLABUS
2025-2026

INSTRUCTOR INFORMATION	FERNANDO LUCAS BACÃO 2 ^o floor, room 10 Phone: 21 3870413 (ext. 222) bacao@novaims.unl.pt http://www.novaims.unl.pt/lbacao IVO BERNARDO ibernardo@novaims.unl.pt VASCO JESUS vjesus@novaims.unl.pt EDUARDO COSTA eduardo.costa@novaims.unl.pt
SCHEDULE	TP2 (P4, P5, P6) Monday 18:30h – 20:15h; TP1 (P1, P2, P3) Tuesday 20:30h – 22:15h;
OFFICE HOURS	Tuesdays at 18h (schedule appointment by email) 2 ^o Floor, Room 10
CONTACT	Students should use Moodle platform for any communication with the course instructors. Homework and project submissions should also be done through Moodle.
DESCRIPTION	The objective of the Data Mining I (DMI) course is to introduce students to the core concepts, methods, and tools of data mining. DMI serves as the first part of a two-part course series, with the second part, offered in the following semester, covering predictive methods in data mining. This course focuses on key paradigms in data mining, including canonical tasks, the nature of inductive learning, and the role of data preparation and preprocessing. It then introduces the major descriptive methods commonly used in the field. While no prior knowledge of data mining is required, students are strongly encouraged to have a basic understanding of inferential statistics and some familiarity with Python. The course strikes a balance between in-depth technical analysis, typical of engineering courses, and managerial perspectives that emphasize the

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Syllabus



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Data Mining I



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10. Semi-supervised classification

9. Clustering techniques

8. Introduction to Cluster analysis

7. Association rules

6. Data preprocessing

1. Introduction to the Data Mining Course

2. Introduction to Data Science

3. The canonical tasks in Data Mining and work process

4. Data visualization

5. Data preparation

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Course Success

COURSE SUCCESS

In this course success depends on a number of factors:

- Basic knowledge of statistics;
- Attend classes;
- Work during the semester and not only when exams are about to start;
- Develop the course project during the semester, making the most of the practical classes;
- Read the suggested references and the PowerPoint slides made available by the lecturer.

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Project

Course Project

The project will be evaluated based on two components. Both components are mandatory:

- **Project Report** (90%). The project consists of the application of the course concepts to a real-world problem using Python. In this project the students will complete the segmentation of a customer database, following all the usual steps of a real-world project. Students will receive a set of specific guidelines that they should follow. The guidelines provide the students with information on the type of tasks they should do and the general results they should achieve. The end-product of the project should be a written report about the database and the different segments of the company, as well as a file with the clustered customers. With this project the students should develop their analytical skills, but also their proficiency in manipulating with large datasets, extracting, transforming and loading tasks, visualization and reporting conclusions.
- **Project Discussion** (10%). After submitting the projects, the students will be called to discuss the project with one of the instructors. Each student will receive an individual grade based on their contribution to the project and discussion.

Project Groups: the project can be done individually or in groups (the latter is a better option) the groups should not exceed 4 students.

Project Deadline: January 3rd. Any delay in the delivery is subject to a penalty of 10% of the grade for each day of delay.

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Grading

Grading

Project: 40%

Exam: 60%

Both components of the evaluation are mandatory. There are two opportunities to do the exam. Any delay in the delivery of the project is subject to a penalty of 10% of the grade for each day of delay. Please note that the project will be developed in groups, but each group cannot have more than 3 elements. To obtain approval in the discipline the student **cannot have less than 8 (40%) in the exam grade.**

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References

Pattern Recognition Letters 31 (2010) 651–666

Contents lists available at ScienceDirect

Pattern Recognition Letters

journal homepage: www.elsevier.com/locate/patrec

ELSEVIER

Data clustering: 50 years beyond K-means^a

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Keywords:
Data clustering
User's dilemma
Methodological developments
Perspectives on clustering
King Bats for justice

ABSTRACT

Organizing data into sensible groups is one of the most fundamental modes of understanding and learning. As an example, a common scheme of scientific classification puts organisms into a system of robust taxo domains, kingdom, phylum, class, etc. Cluster analysis is the formal study of methods and algorithms for grouping, or clustering, objects according to measured or presumed intrinsic characteristics or similarity. Cluster analysis does not use category labels that tag objects with prior definitions, i.e., class labels. The absence of category information distinguishes data clustering (unsupervised learning) from classification or discriminant analysis (supervised learning). The act of clustering is its own structure in data and is therefore explanatory in nature. Clustering has a long and rich history in a variety of scientific fields. One of the most popular and simple clustering algorithms, K-means, was first published in 1955 in spite of the fact that K-means was proposed over 50 years ago and thousands of clustering algorithms have been published since then. It remains in wide self-use today. This article reviews the difficulty in designing a general purpose clustering algorithm and the fit-point problem of clustering. We provide a brief overview of clustering, summarize well-known clustering methods, discuss the major challenges and key issues in designing clustering algorithms, and present our view of the emerging and useful research directions, including semi-supervised clustering, ensemble clustering, simultaneous feature selection during data clustering, and large scale data clustering.

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1. Introduction

Advances in sensing and storage technology and dramatic changes in applications such as Internet search, digital imaging, and video surveillance have created many high-volume, high-dimensional data sets. It is estimated that the digital volume of stored information increased approximately 281 exabytes in 2007, and it is projected to reach 130 exabytes by the year 2011 (Exabyte = 10^{18} bytes, or 1,000,000 terabytes) (Carrico, 2008). Most of the data is stored digitally in electronic media, thus providing huge potential for the development of automatic data analysis, classification, and retrieval systems.

streams are unstructured, adding to the difficulty in analyzing them.

The increase in both the volume and the variety of data requires advances in methodology to automatically understand, process, and summarize the data. Data analysis techniques can be broadly classified into two major types (Takis, 1977): (i) exploratory or descriptive, meaning that the investigator does not have pre-specified models or hypotheses but wishes to understand the general characteristics or structure of the high-dimensional data, and (ii) confirmatory or inferential, meaning that the investigator wants to confirm the validity of a hypothesis/hypotheses or a set of assumptions.

Data Clustering: A Review

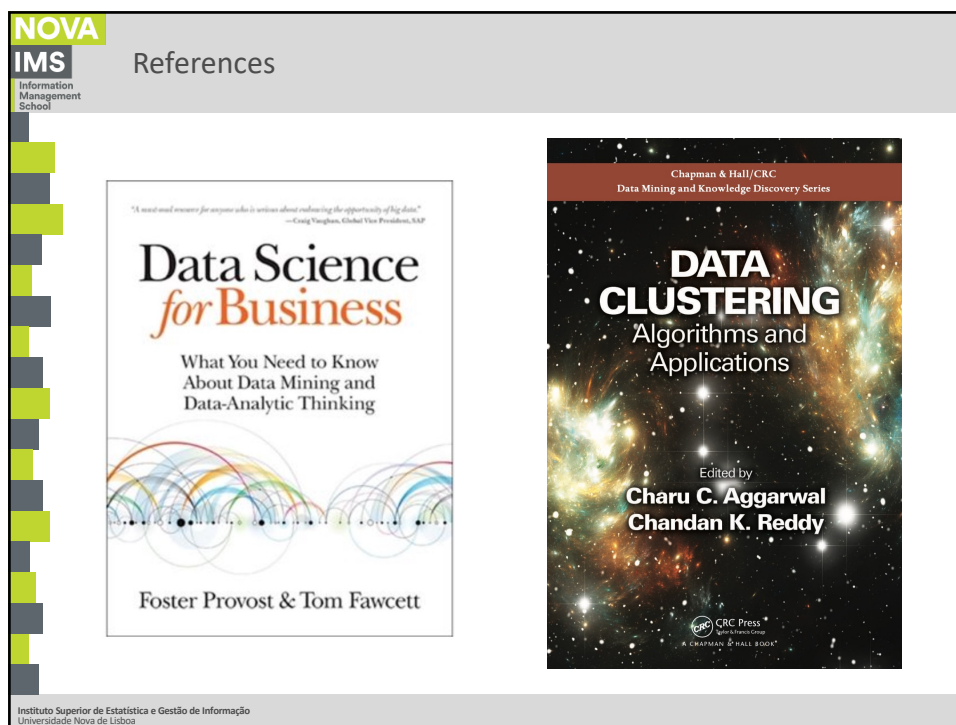
A.K. Jain
Michigan State University
and
M.N. Murty
Indian Institute of Science
and
P.J. Flynn
The Ohio State University

Clustering is the unsupervised classification of patterns (observations, data items, or feature vectors) into groups (clusters). The clustering problem has been addressed in many contexts and by researchers in many disciplines; this reflects its broad appeal and usefulness as one of the steps in exploratory data analysis. However, clustering is a difficult problem combinatorially and differences in assumptions and contexts in different communities has made the transfer of useful generic concepts and methodologies slow to occur. This paper presents an overview of pattern clustering methods from a statistical pattern recognition perspective, with a goal of providing useful advice and references to fundamental concepts accessible to the broad community of clustering practitioners. We present a taxonomy of clustering techniques and identify cross-cutting themes and recent advances. We also describe some important applications of clustering algorithms such as image segmentation, object recognition, and information retrieval.

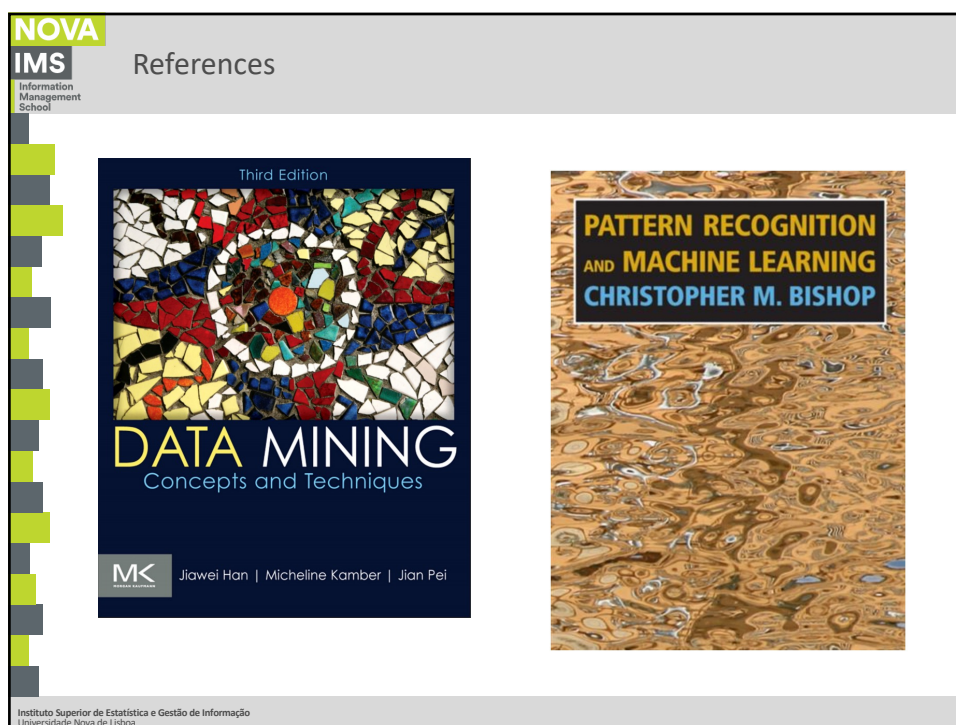
Categories and Subject Descriptors: I.5 [Computer Methodologies]: Pattern Recognition—Models; I.5.3 [Pattern Recognition]: Clustering; I.5.4 [Pattern Recognition]: Applications—Computer Vision; H.3.3 [Information Storage and Retrieval]: Information Search and Retrieval—Clustering; I.2.6 [Artificial Intelligence]: Learning—Knowledge Acquisition

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Agenda

- The Growth of the Digital Universe
- Buzz Words and Definitions
- Data Science
- Different Roles in Data Science
- Data-Driven Thinking
 - The benefits of data-driven thinking
 - Statistics vs Data Science
 - How to develop models?
 - Building Features

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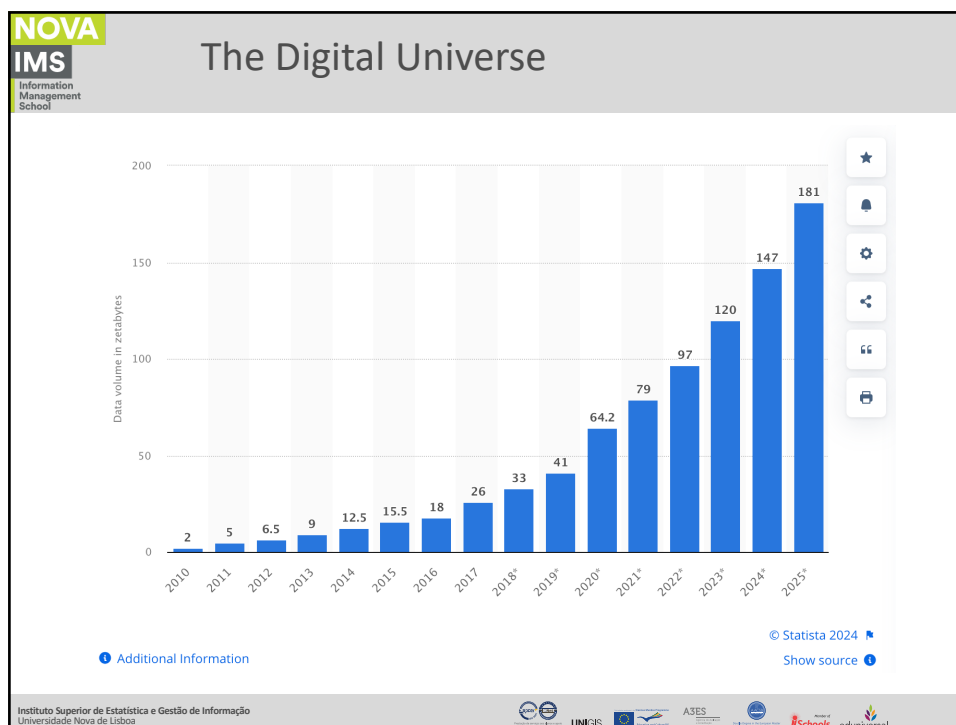
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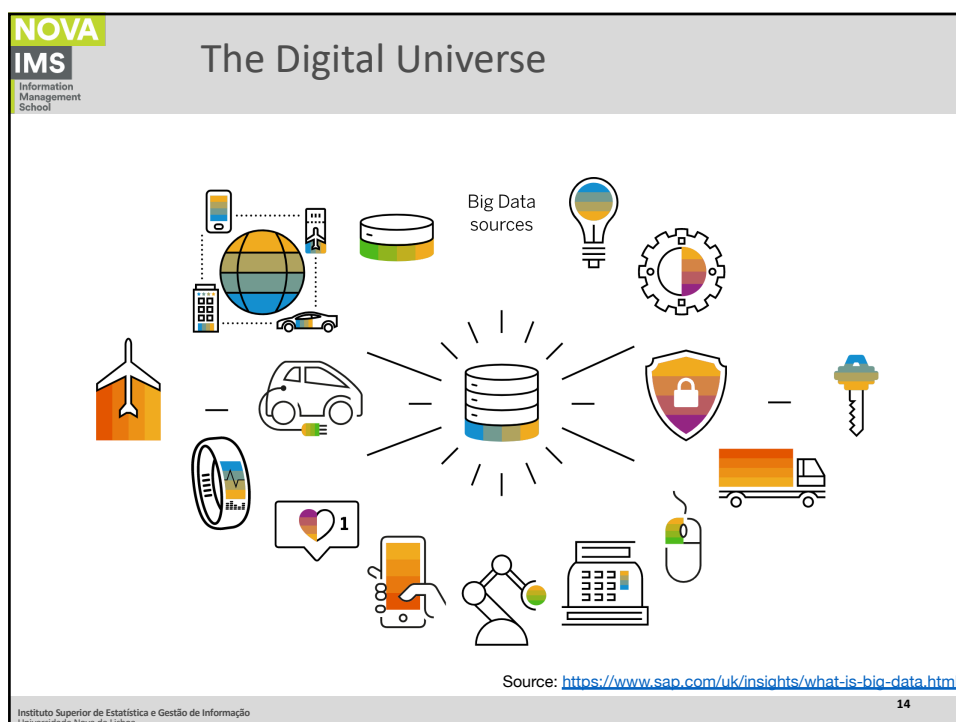
The Growth of the Digital Universe

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The Digital Universe



The Apollo 11 lunar mission which landed on the moon used a computer that was 1,300 times less powerful than the iPhone

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The Digital Universe



- The current A350 model has a total of close to 6,000 sensors across the entire plane and generates 2.5 Tb of data per day, while the newer models will capture more than triple that amount.

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The Data Gap

“Data have swept into every industry and business function and are now an **important factor of production, alongside labor and capital.”**

McKinsey Global Institute

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The Data Gap

“Although the volume of data created is expected to **increase fivefold** over the next five years, best-guess estimates suggest that **less than 10 percent** of the information created is meaningfully organized or deployed. That number will only shrink as the rate of information production goes up.”

“The productivity imperative”, Bisson, P., Stephenson, E., Viguerie, S., McKinsey Quarterly June 2010)

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“Information streams are the infinite by-product of a knowledge economy—the best companies will turn this free good into gold”

“A final productivity driver will be something businesses are creating in digital bucket loads: information.”

“The productivity imperative”, Bisson, P., Stephenson, E., Viguerie, S., McKinsey Quarterly June 2010)

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The Data Gap

- Telemetry - a host of electronic devices, including ECU (Engine Control Unit or as some people call it Electronic Control Unit) which transmits specific data, for example measurements, but not only, to a remote site, in F1 case, to pit wall and pit garage. It electronically records performance of engine, status of suspensions, gearbox data, fuel status, all temperature readings including tires temperature, g-forces and actuation of controls by the driver. The data is then used as a foundation for determining car setup and all problems.

By the Numbers

PER-LAP DATA
100 individual sensors on the car produce 20-30 MB of data, of which 3-4 MB are sent to the pit wall platform in real-time, including information about the time, engine, temperature, and fuel usage, which is then used by the pit team to make immediate decisions about in-race strategies.

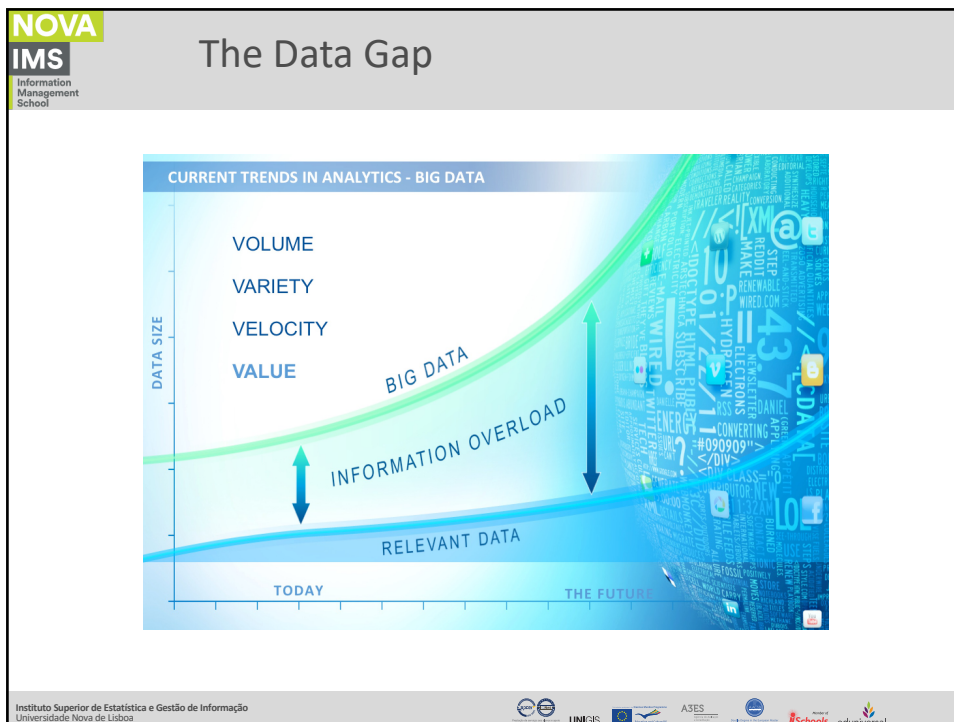
TELEMETRY
data collected 25GB per three-day race weekend
Data transmitted to Sauber HQ in Hanoi, Switzerland, during an hour of free practice, 600MB

A Fraction of a Second can be the difference between QUALIFYING OR NOT in the top 10. During any given race, qualifying times of the first 15 cars were likely to be within a span of just 1 sec.

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P&G - "Business Spheres"



<https://hbr.org/2013/04/how-p-and-g-presents-data>

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The Economist: The data deluge



<http://www.economist.com/node/15579717>

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Buzz Words and Definitions

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Buzzwords

“Data Mining!”

“Predictive!”

“Advanced Analytics!”

“Data Science!”

“Machine Learning!”

“Cognitive!”

“Artificial Intelligence!”

TimoElliott.com

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Big Data

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Big Data

***‘we will simply take big data to mean
datasets that are **too large for
traditional data-processing systems**
and that therefore require new
technologies.’***

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Big Data





- **Volume:** While volume is by no means the only component that makes Big Data “big,” it is certainly a **primary feature**. To fully manage and utilise Big Data, advanced algorithms and AI-driven analytics are required.
- **Velocity:** In the past, any data that was generated **had to later be entered into a traditional database system – often manually** – before it could be analysed or retrieved. Today, Big Data technology allows databases to process, analyse, and configure data while it is being generated – sometimes within milliseconds.
- **Variety:** Data sets that are comprised solely of structured data are not necessarily Big Data, regardless of how voluminous they are. Big Data is typically comprised of combinations of structured, unstructured, and semi-structured data. Traditional databases and data management solutions lack the flexibility and scope to manage the complex, disparate data sets that make up Big Data.

Source: <https://www.sap.com/uk/insights/what-is-big-data.html>

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Big Data

Key Spotify Statistics

551 monthly active Spotify users;
220 million Spotify Premium subscribers;
25 hours spent listening per month;
Average revenue per user \$4.72
82 million tracks available on Spotify;

<https://www.businessofapps.com/data/spotify-statistics/>

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The Data Deluge

“The number of **songs on Spotify** jumped from **70m to 82m in 2021**, or **33k each day**.

Notably, **1.5% of songs had 500k+ lifetime streams**, and **25.8% had 100 streams or fewer**.

Assuming the average song on Spotify runs 3 minutes, 30 seconds, it would take **~546 years to listen to all of them.**”

Source: <https://thehustle.co/4-18-22-digits>

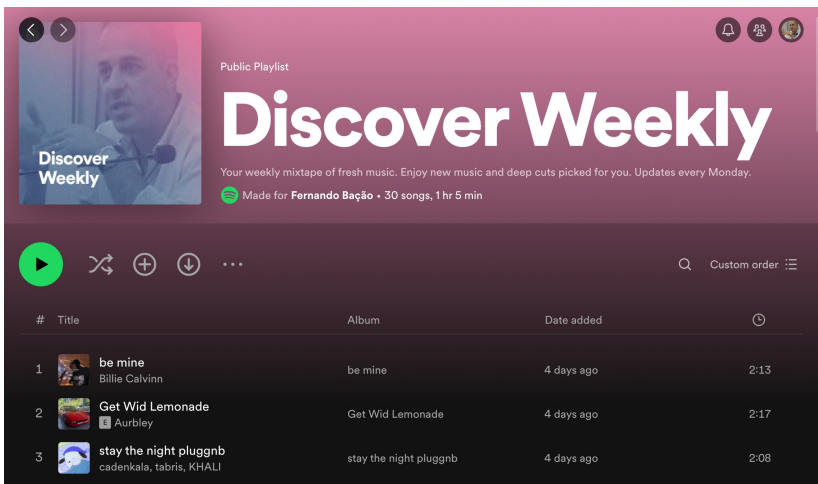
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Big Data



#	Title	Album	Date added	
1	be mine Billie Calvinn	be mine	4 days ago	2:13
2	Get Wid Lemonade Aurbley	Get Wid Lemonade	4 days ago	2:17
3	stay the night pluggnb cadenkala, tabris, KHALI	stay the night pluggnb	4 days ago	2:08

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Artificial Intelligence

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Artificial Intelligence

- Artificial Intelligence is the science of making things smart, and can be defined as:
“human intelligence exhibited by machines”
- A broad term for **getting computers to perform human tasks**
- The **scope** of AI is disputed and constantly **changing over time**

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Artificial Intelligence

- The systems implemented today are a form of narrow AI – a system that can do **just one (or a few)** defined things as well or **better than humans**
 - Like recognizing objects
 - Play chess
 - Jeopardy

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Machine Learning

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Machine Learning

- Machine learning can be defined generally as:
 - An **approach to achieve Artificial Intelligence** through systems that **can learn from experience to find patterns on a set of data**
 - ML involves **teaching a computer to recognize patterns by example**, rather than programming it with specific rules.
 - These patterns can be **found within data**

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Machine Learning

- Machine learning can be defined generally as:

ML is about creating algorithms (or a set of rules) that learn complex functions (or patterns) from data and make predictions on it

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- Machine learning can be defined generally as:
- ML is essentially about predicting stuff
 1. It takes some **data, to train** the system
 2. **Learns patterns** from this data
 3. **Classifies new data** it has not seen before (makes a guess on what probably is based on the knowledge gained in the previous step)

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Machine Learning

- What is extremely cool about ML is that it learns by itself from the data

Write a computer program with **explicit rules** to follow

```
if email contains V!agrå
then mark is-spam;

if email contains ...

if email contains ...
```

Traditional Programming

Write a computer program to **learn from examples**

```
try to classify some emails;
change self to reduce errors;
repeat;
```

Machine Learning Programs

FEUP

FEUC

FEIS

FEUP

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Machine Learning

- What is extremely cool about ML is that it learns by itself from the data

Traditional Programming

Data

Program

Computer

Output

Machine Learning

Data

Output

Computer

Program

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Artificial Intelligence

Machine Learning

1950's 1960's 1970's 1980's 1990's 2000's 2010's

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Artificial Intelligence vs Machine learning

Under the bonnet
How a self-driving car works

The diagram illustrates a self-driving car with various sensors and their functions:

- GPS (global positioning system):** Signals from GPS satellites are combined with readings from tachometers, altimeters, and gyroscopes to provide more accurate positioning than is possible with GPS alone.
- Lidar (light detection and ranging):** sensors bounce pulses of light off the surroundings. These are analysed to identify lane markings and the edges of roads.
- Video cameras:** detect traffic lights, read road signs, keep track of the position of other vehicles and look out for pedestrians and obstacles on the road.
- Radar sensor:** The information from all of the sensors is analysed by a central computer that manipulates the steering, accelerator and brakes. Its software must understand the rules of the road, both formal and informal.
- Ultrasonic sensors:** may be used to measure the position of objects very close to the vehicle, such as curbs and other vehicles when parking.
- Radar sensors:** monitor the position of other vehicles nearby. Such sensors are already used in adaptive cruise-control systems.

Source: The Economist

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Artificial Intelligence and Machine Learning

The diagram illustrates the hierarchy of Artificial Intelligence and Machine Learning:

- Artificial Intelligence** (outermost layer)
 - Machine Learning**
 - Deep Learning**
 - Foundation Models**
 - LLM, Vis., Aud.
 - Generative AI
- Other AI models:** LG, CT, NN, Ct, SVM

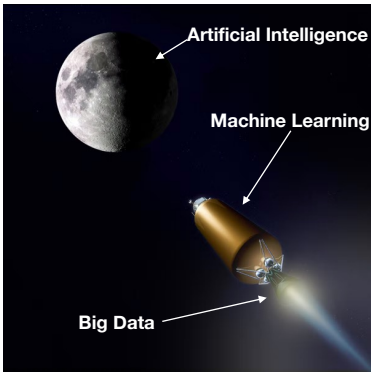
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Artificial Intelligence and Machine Learning



Source: AI Luminary Series: Pedro Domingos Explains Machine Learning <https://www.intel.com/content/www/us/en/analytics/ai-luminary-pedro-domingos-video.html>

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Questions?

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